

Towards Multi-Person Gesture Recognition using Commodity Wi-Fi

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Abstract—Comparing the recognition of human gestures using cameras, radar, or LiDAR, a Wi-Fi-based gesture recognition system has distinct advantages, such as being low-cost, being device-free, and having much less privacy leakage. Recently, there have been advancements in Wi-Fi-based gesture recognition, but most of the research has primarily focused on single-person gesture recognition. However, in real-world scenarios like e-learning, it is common for multiple individuals to engage in different actions simultaneously. To this end, this paper focuses on multi-person gesture recognition, which presents two major challenges, that is, the recognition accuracy due to the Wi-Fi signal interference, and the processing time for real-time applications. Multi-person gesture recognition is more challenging than single-person scenario due to the interference caused by the superposition of Wi-Fi signals induced by multiple moving individuals. In this paper, we define a concept of super-gesture and propose a Wi-Fi-based Super-Gesture recognition (WiSG) method. Through the decomposition of the super-gesture's DFS spectrogram by Multi-Motion Trajectory algorithm, we extract modified signals of each person. Moreover, a novel feature called Field Motion Velocity is proposed by fully exploiting the advantages of our multiple transmitter-receiver Wi-Fi sensing system. The proposed feature is not significantly affected by domains such as position, orientation, and other factors irrelevant to gestures. As a result, our approach can effectively recognize gestures across different domains. Evaluation results show that the cross-domain recognition accuracy of our WiSG can achieve up to 89% in multi-person scenario. Moreover, our approach can reduce processing time by 20 times against Widar3.0, which satisfies the requirements of most real-time applications.

Index Terms—Gesture Recognition, Wi-Fi Sensing, Cross-domain Effect, Multi-person Gesture Perception

I. INTRODUCTION

The recognition of human gestures is a crucial but challenging component in various IoT applications that involve human participation, such as e-learning [1], assisted living [2], e-health [3], and virtual reality [4]. Conventional approaches often use cameras (2D images) [5], [6], radars (depth maps) [7], [8], or LiDAR (3D point clouds) [9], [10] to recognize human gesture. Camera-based methods are effective and low-cost due to the advancement of computer vision techniques, but the privacy concerns like exposing human face greatly prevent their deployment in many privacy-sensitive applications. Radar/LiDAR-based methods typically need dedicated hardware and are too expensive and power-consuming, while without exposing human face.

Recently, RF-sensing based methods [11], [12] have been proposed and have attracted increasing research interest. One

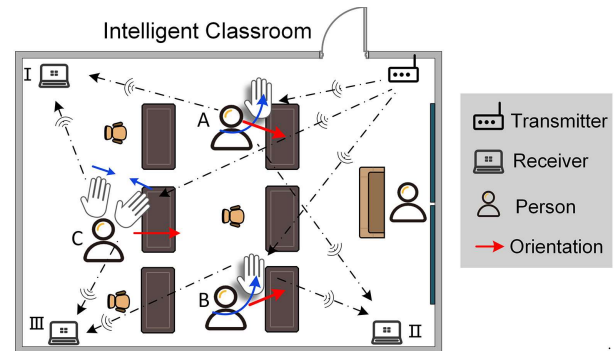


Fig. 1: An example for multi-person scene. In classroom setting, it is typical for multiple students to perform various actions simultaneously (e.g., raising/lowering hand or waving hand). To precisely recognize these concurrent gestures, we build a Wi-Fi sensing system that uses multiple transmitters and receivers (e.g., 1 transmitter and 3 receiver).

representative method is to use Wi-Fi signals to recognize human activity [13]–[20]. Wi-Fi devices are a cost-effective alternative to radar/LiDAR-based approaches, as they are commonly used in homes, hospitals, and classrooms. Additionally, Wi-Fi technology poses no risk of privacy leakage.

Even with these advantages, Wi-Fi-based methods still face a major technical challenges. The Wi-Fi signals often carry adverse environmental factors that are unrelated to human gesture, including the *position* and *orientation* of the human, as well as the *environment* that affects the propagation of Wi-Fi signals, such as the layout of furniture and walls. One configuration of all these factors is called a *domain* [18]. The Wi-Fi signals for the same gesture but in different domains exhibit significant differences, which is referred to as the *cross-domain* effect, as shown in Fig. 2. When a gesture recognition model is trained on one specific domain, such as a restroom, and then applied directly to another domain, such as a classroom, its recognition performance will degrade significantly. Therefore, improving cross-domain recognition accuracy is a critical challenge in Wi-Fi-based gesture recognition.

To address this challenge, certain methods utilize transfer learning [21] and adversarial learning [22] techniques. One example is the method proposed by CrossSense [21], which utilizes a Mixture-of-Experts (MOE) approach to extract domain-independent features from the correlation between

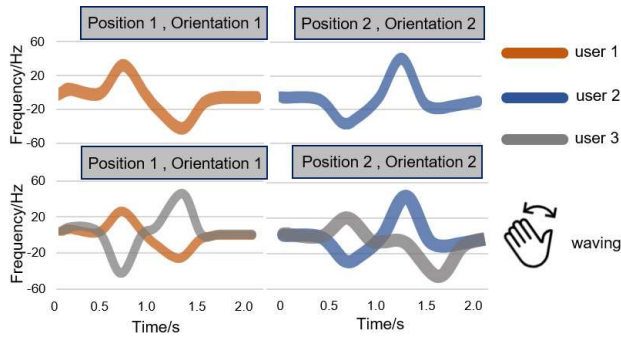


Fig. 2: The cross-domain effect. (1) first-row are single-person scenarios, where spectrogram changes for the same gesture (*i.e.*, waving) in different domains (*e.g.*, position, orientation). (2) second-row are two-person scenarios, where spectrogram changes drastically against single-person scenarios.

various WiFi inputs and outputs. Different from these machine learning-based approaches, another solution, Widar3.0 [18], attempts to solve such a problem at the lower signal level, rather than the upper model level. Specifically, they propose a method to extract domain-independent features reflecting only gesture itself from raw domain-dependent WiFi signals. However, it can only handle human position and orientation factors and still suffers from other environmental factors.

On the other hand, the majority of current studies concentrate on *single-person* gesture recognition, which assumes that there is only one person present in the room performing an action. But for realistic scenarios (*e.g.*, classrooms, hospitals, offices, etc.), there usually are more than one person staying in the room, and they may take different actions simultaneously. Therefore, how to recognize those *concurrent gestures from multiple person* is very important, as shown in Fig. 1.

Multi-person gesture recognition is more challenging than single-person scenarios. The reason is that WiFi signals, which are reflected by the gestures of different individuals, will cause interference with one another. Various person take different gestures, resulting in a more intricate WiFi signal compared to that of a single person, as illustrated in Fig. 4. Moreover, beyond the traditional domain factors (position, orientation, and environments), a new factor of the spatial relation among multiple people is introduced for the multi-person scenario. Thus, the cross-domain effect becomes more serious for multi-person scenarios than single-person, as shown in Fig. 2.

Note that the previous cross-domain methods for single-person scenarios cannot be directly adopted to the multi-person scenario. For example, in Widar3.0, the BVP feature is dedicated designed for single-person scenarios, *i.e.*, the coordinate system is tied to the single person. Regarding a multi-person scenario, even if we can tie a distinct coordinate system to each individual, the BVP features for distinct persons - although they are taking the same action - will differ from each other due to the un-calibrated coordinate systems. Our empirical results in the experimental section will illustrate it.

In this paper, we focus on the problem of multi-person

gesture recognition. In order to recognize concurrent gestures from multiple persons, we regard multiple persons as one whole object, where each person is treated as a part of the object. Thus, those concurrent gestures can be regarded as a *super-gesture* (SG), which comes from the whole object, and each gesture can be regarded as a *sub-gesture*, which comes from an object part. In other words, one super-gesture is a combination of multiple sub-gestures. In this way, the recognition of multiple gestures can be cast as the recognition of one super-gesture, so that the method for single gesture recognition can be adapted to solve this problem.

However, the recognition of super-gestures is more complex than that of single gestures. Particularly, the DFS spectrogram for a super-gesture is the composition of many DFS spectrograms corresponding to different sub-gestures. The DFS spectrogram for a super-gesture is much more complex than that for a single gesture. Therefore, it is difficult to use single gesture recognition methods directly to recognize super-gestures. For example, if we have five people $a_1, a_2, \dots, a_5$ in the room, each person can perform two gestures: raising/lowering hand (h) or waving hand (w). Thus, we can define 5×2 different super-gestures, *e.g.*, $S_1 : (a_1 = h, a_2 = w, a_3 = h, \dots, a_5 = w)$. Considering two super-gestures, where four person perform the same gesture but only one person performs a different gesture, the DFS difference for the two super-gestures is too small to be effectively distinguished. Secondly, improving cross-domain recognition becomes more challenging for super-gesture than single gesture recognition.

To this end, we propose a WiFi-based Super-Gesture recognition (WiSG) approach, the workflow of which is shown in Fig. 3. Our main contributions are twofold: (1) Regarding the complex DFS spectrogram for a super-gesture, we avoid directly recognizing it, but propose to decompose it into several DFS components such that each component corresponds to a sub-gesture. Specifically, inspired by the seam carving mechanism [23], we design a *Multi-Motion Trajectory* (MMT) module which can disentangle those DFS components as much as possible. (2) Secondly, to mitigate the cross-domain effect, we propose a *Field Motion Velocity* (FMV) module by fully exploiting the advantages of our multiple transmitter-receiver WiFi sensing system. If we have three transmitters and three receivers, we will have 3×3 WiFi propagation links. For each link, we can obtain a DFS spectrogram feature. Although all these DFS features contain information about the human gesture, if they are simply fused together, the final recognition performance cannot be effectively improved, since these features are not well aligned. But in our FMV module, we project all DFS spectrograms into the same coordinate system and summarize them over all links. With our delicate design, those DFS features can be fully exploited, and furthermore, the produced features are not significantly affected by domain, which effectively mitigates the cross-domain effect.

We use our approach to evaluate both single-person and multi-person scenarios. For the single-person scenario, our approach achieves above 90% cross-domain recognition accuracy, which is comparable to the state-of-the-art single-

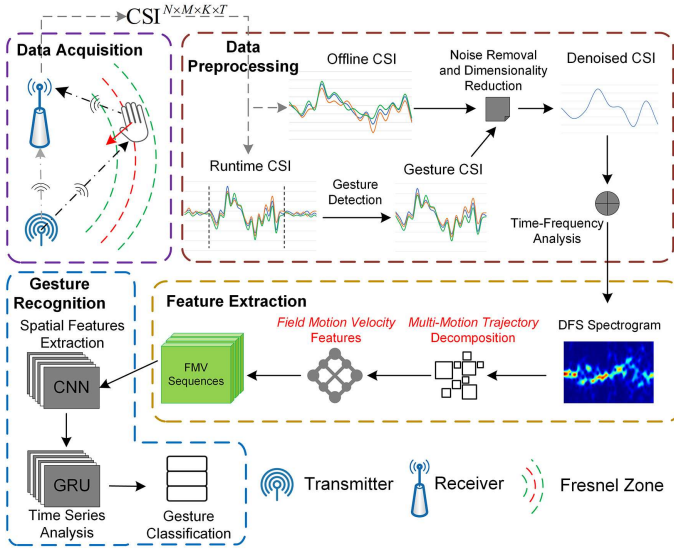


Fig. 3: The workflow of our WiSG.

person solution (*i.e.*, Widar 3.0). For multi-person scenarios, a baseline method is to directly adapt Widar 3.0 to multi-person scenario. Due to neglecting the interference among different individual's gesture, its performance is degraded significantly. In particular, our approach achieves 89% cross-domain recognition accuracy, which has a 10% improvement against the baseline method. Besides, our approach could reduce processing time by 20 times against baseline method, which satisfies the requirements of most real-time applications.

Our main contributions are summarized as follows:

- We propose a method for multi-person gesture recognition using WiFi. We define a concept of super-gesture and propose a MMT module to decompose its DFS spectrogram via seam carving mechanism.
- We propose an FMV module to fully exploit the advantages of our multiple transmitter-receiver WiFi sensing, which can effectively mitigate the cross-domain effect.

II. OVERVIEW OF WiSG

We propose WiSG, a multi-person gesture recognition system with the ability of cross-domain using commodity WiFi. The system consists of three modules: data acquisition and preprocessing, feature extraction, and gesture recognition. The workflow of WiSG is shown in Fig. 3.

Data acquisition and preprocessing. The objective of this module is to remove noise from the received signal and decrease its size for subsequent feature extraction. Specifically, with time-frequency analysis, we obtain the power spectrum in both time and Doppler frequency domains, which is called the DFS spectrogram. For real-time applications, we also have a gesture detection algorithm to find time segments which contain human gestures. Once a gesture is detected, the subsequent modules will identify its type.

Feature Extraction. Given the previously calculated DFS spectrogram, we apply the MMT algorithm to decompose the DFS spectrogram and obtain DFS components corresponding

to different sub-gestures. Then, the MMT features are divided into multiple equal-length slices along the time, and our proposed FMV module is used to build a domain-independent feature with the ability of cross-domain recognition.

Gesture Recognition. This module utilizes a hybrid neural network that combines CNN and GRU to classify gestures. The first step is normalization, which eliminates the adverse effects caused by possible outlier data. Next, the normalized FMV features are inputted into a CNN network for spatial modeling. Finally, a GRU network is utilized for temporal modeling, which produces the final recognition results.

III. DATA ACQUISITION AND PREPROCESSING

Data acquisition and preprocessing involve processing the received CSI (Channel State Information) and building their representations. Following previous methods, we perform time-frequency analysis and obtain the power spectrum in time and frequency domains, which is called the DFS spectrogram.

For real-time applications, a gesture detection algorithm is proposed to find time segments, which contain human gestures. Specifically, we adopt a threshold-based gesture detection algorithm to detect the beginning and end of a gesture, and only perform gesture recognition during the time segments when the gesture is active, which can significantly reduce resource consumption and system load. The first principal component of PCA is used for gesture detection. We observe that the amplitude fluctuates considerably when there is gesture activity in the surrounding environment. Conversely, when there is no gesture activity, the amplitude remains relatively stable. To detect gestures, we use a sliding window to calculate the variance of CSI amplitude at each time point and use this variance as a threshold for detection.

IV. FEATURE EXTRACTION

Given the previously calculated DFS spectrogram, we need to build a domain-independent feature for the super-gesture. The first step is to decompose the DFS spectrogram into several DFS components, where each component corresponds to a sub-gesture. The second step is to build a domain-independent feature with the ability of cross-domain recognition.

A. Mult-Motion Trajectory Decomposition

The previously calculated DFS spectrogram corresponds to a super-gesture, while we want to obtain DFS spectrograms corresponding to different sub-gesture. Previous studies [15] show that the superposition of signals from multiple gestures is a linear combination, which does not change the frequency of each gesture. When two person perform gestures simultaneously, they have two frequency components in the power spectrum, respectively. As shown in Fig. 4, each gesture corresponds a continuous trajectory in the DFS spectrogram, and multiple gestures show the superposition of multiple continuous trajectories. Therefore, to recognize the concurrent gestures, we decompose the DFS spectrogram into distinct DFS components, which corresponds to different sub-gesture.

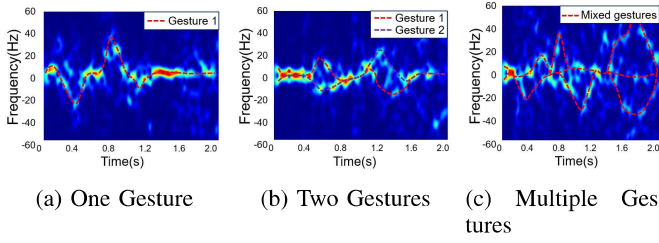


Fig. 4: The DFS spectrogram for a super-gesture contains several components, where each component corresponds to a sub-gesture.

However, due to the superposition of power components in the DFS spectrograms caused by different gestures, we need to establish a quantitative relationship between different gestures and power components. Secondly, we also need to remove irrelevant interference in the DFS spectrogram.

To this end, we propose a Multi-Motion Trajectory (MMT) decomposition algorithm based on the seam carving mechanism [23]. To establish a quantitative relationship between power components and the area of reflection regions, we assume that there are K body parts performing gestures and dominating at time t . We model the power matrix $P_d(t)$ of Doppler shift f_D in the DFS spectrogram at time t using Eq. (1a) by introducing an acceptable computational error σ . As explained in CARM [15], the power distribution of the spectrogram for a specific Doppler shift at time t vary with the change of the reflection area S . Therefore, we introduce a conversion factor θ to establish the relationship between $P_d(t)$ and the k -th reflection area $S_d(t, k)$, as shown in Eq. (1b):

$$P_d(t) = \left\{ \sum_{k=1}^K P_d(t, k) \cdot \text{rect}\left(\frac{f_{ds}(t, k) - f_{D_i}}{2\sigma}\right) \right\}_{i=1}^n \quad (1a)$$

$$= \theta \left\{ \sum_{k=1}^K S_d(t, k) \cdot \text{rect}\left(\frac{f_{ds}(t, k) - f_{D_i}}{2\sigma}\right) \right\}_{i=1}^n, \quad (1b)$$

where $P_d(t, k)$ and $f_{ds}(t, k)$ represent the power and frequency shift generated by the movement of the k -th body part at time t , respectively, n represents the number of frequency shift components in the DFS spectrogram, f_{D_i} represents the i -th frequency shift component in the DFS spectrogram, and $\text{rect}(\cdot)$ denotes a matrix function.

However, due to the difficulty in estimating the resolution of WiFi signals and the number of body parts in motion, we introduce propagation loss c to obtain an approximate value of power matrix $\hat{P}_d(t)$ at time t :

$$\hat{P}_d(t) \approx \theta \left\{ c \sum_{k=1}^K S_d(t, k) \cdot \text{rect}\left(\frac{f_{ds}(t, k) - f_{D_i}}{2\sigma}\right) \right\}_{i=1}^n. \quad (2)$$

The final total DFS power matrix and the quantitative relationship with the area of each body part can be represented as $P_d = \left\{ \hat{P}_d(t) \right\}_{t=1}^T$. There is a relationship between human motion speed v and Doppler frequency shift f_D , which is expressed as $f_D = -\frac{v}{\lambda}$ according to Widar2.0 [24], where

Algorithm 1 Multi-motion Trajectory Decomposition Algorithm.

Input: $P_{f_D \times T \times R}$, T_s , K , Attenuation factor Att .

Output: $MMT_{f_D \times T_s \times R}$.

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1: Initialize  $MMT_{f_D \times T_s \times R}$ ;
2: for each receiver  $r \in 1, 2, \dots, R$  do:
3:    $P_{seg_{f_D \times T_s}} = \text{SegmentMean}(P_{f_D \times T \times R}(r), T_s)$ ;
4:   Initialize multiple paths  $Path\_sum_K$ ;
5:   for each path  $k \in 1, 2, \dots, K$  do:
6:      $energy_{f_D \times T_s} = \text{Gradient}(P_{seg_{f_D \times T_s}})$ ;
7:      $s_k^* = DP\left(\max\left(\sum_{i=1}^{T_s} energy_{f_D \times T_s}(i)\right)\right)$ ;
8:      $Path\_sum_K = Path\_sum_K + s_k^*$ ;
9:      $P_{seg_{f_D \times T_s}} =$ 
        $PathAttenuation(P_{f_D \times T_s}, s_k^*, Att)$ ;
10:    Update energy value;
11:  end for;
12:   $MMT_{f_D \times T_s \times R}(r) = Path\_sum_K$ ;
13: end for;
14: return  $MMT_{f_D \times T_s \times R}$ ;

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λ represents the wavelength of the signal. Therefore, we can use frequency shift changes instead of motion speed changes. Next, in order to accurately extract multi-motion trajectory, we need to remove irrelevant interference from the DFS spectrogram and carve out multi-motion trajectories for multiple moving body parts with time continuity. Inspired by seam carving mechanism in image processing, we designed a MMT decomposition algorithm based on DFS spectrogram.

Specifically, we first assign an energy value to each power in the DFS spectrogram. For any time t and any frequency f , we define the energy matrix as $energy(\hat{P}_d(f, t)) = \left| \frac{\partial}{\partial f} \hat{P}_d(f, t) \right| + \left| \frac{\partial}{\partial t} \hat{P}_d(f, t) \right|$ using gradients. Then, we apply dynamic programming to find a horizontal eight-connected path with maximum energy (i.e., optimal path), defined as $s^* = \max_s \sum_{i=1}^n energy(\hat{P}_d(i, t))$, where n represents the number of frequency shift components. We then add this optimal path to our MMT feature and finally find the top K dominant paths. Since motion of individual body part is continuous, while that of multiple body parts lacks continuity, different trajectories for motion of individual body part can be carved out. To address the problem of overlapping power values caused by arm movements with identical speed components, we halve the power value of the optimal path and repeat generating it until finding K optimal paths as MMT features.

The entire process for extracting MMT features is described in Algorithm 1. Note that selecting an appropriate attenuation factor Att is important: if Att is too small, it may not be possible to extract multiple body parts with identical speed components smoothly; if Att is too large, non-existent body parts may be extracted instead. After conducting numerous experiments and validations, an optimum Att value was found to be 0.5. T_s (the number of time segments) selection is also crucial because if a time segment lasts too long or too short, the feature may become too average or lose motion informa-

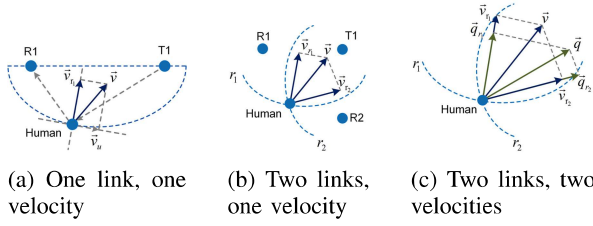


Fig. 5: Motion velocity projection based on the Fresnel zone.

tion. After experimental verifications, the optimal duration for each time segment was set to 30ms, and the number of time segments is calculated as $T_s = \frac{T}{30ms}$. The final dimension of MMT features generated is $f_D \times T_s \times R$, which are then used to construct an FMV model for generating FMV features.

B. Field Motion Velocity Features

Due to the superposition of multiple gesture signals, even though we have separated multiple trajectories in Section IV-A, all existing cross-domain recognition solutions are still unable to achieve cross-domain recognition of multiple gestures. For example, Widar3.0 proposed BVP features to achieve cross-domain recognition, but BVP features are highly correlated with the position and orientation of a single user and cannot be extended to multi-person scenario. To solve this problem, we propose a Field Motion Velocity (FMV) model based on the Fresnel zone model [25] that allows for super-gesture recognition by adding different motion velocity components received by different WiFi links into the entire field coordinate system, overcoming correlation issues.

Due to the Fresnel zone effect, as shown in Fig. 5a, arm movement generates a velocity $\vec{v}$, and the CSI dynamic component generated on the i -th transceiver link by $\vec{v}$ is caused by the velocity component $\vec{v}_{r_i}$, whose orientation is normal to an ellipse with the i -th transmitter and receiver as foci. Therefore, the CSI dynamic components of multiple transceiver links are caused by velocity components on different normal orientations, as shown in Fig. 5b. Two velocities of equal magnitude but different orientations, $\vec{v}$ and $\vec{q}$, decomposed along two WiFi links are shown in Fig. 5c. We observe that the sum of their speed components on both WiFi links is very close. The basic idea of FMV model is based on this observation to fuse motion information generated from different WiFi links. The purpose of fusion is to make identical super-gestures have similar features across domains and ultimately build a cross-domain model for super-gesture recognition.

In this experiment, we convert the frequency shift of MMT features from different transceiver links into velocity and decompose it into a two-dimensional coordinate system in the field that we have constructed. Section IV-A describes the relationship between human movement speed v and frequency shift f_D as $f_D = -\frac{v}{\lambda}$. At CH-165, for the 5.825GHz WiFi signal, a DFS resolution of 1Hz corresponds to only 0.05m/s of motion change. Due to uncertainty principle, the frequency resolution of DFS power spectrum is only about 10Hz. Considering that the range of frequency shifts generated

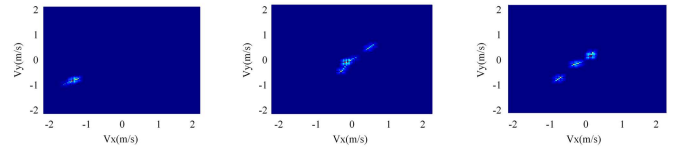


Fig. 6: FMV features for different gestures.

by human movement is within ± 60 Hz [18], one link's DFS power spectrum can only provide about 12 velocity constraints. Taking all factors into consideration, in our constructed field velocity model, we set the velocity range and resolution along x-axis and y-axis to be ± 2.4 m/s and 0.24m/s, respectively. Thus, each axis provides 20 sampled velocities in our two-dimensional coordinate system for the field domain ultimately.

Specifically, we establish a two-dimensional coordinate system, with the x and y axes aligned with the orientations of transceiver links I and II, respectively. Note that in the experimental setup, transceiver link-I are perpendicular to transceiver link-II. For a single user, their position is taken as the origin of the field's two-dimensional coordinate system. For multiple users, the maximum x and y values of each person's distance from the transmitter are taken as separate origins for the field's two-dimensional coordinate system. At each time t , there exists a velocity matrix of dimensions $M \times M$ to represent velocities across the field, where M represents possible velocities along each axis. The following derivation applies only to path k in MMT features. Assuming that transmitter-receiver positions for path i are $\mathbf{l}_t^{(i)} = (x_t^{(i)}, y_t^{(i)})$, $\mathbf{l}_r^{(i)} = (x_r^{(i)}, y_r^{(i)})$, with current target position at origin, any arbitrary velocity $\mathbf{v} = (v_x, v_y)$, then speed components for path i can be expressed as:

$$v^{(i)} = a_x^{(i)} \cdot v_x + a_y^{(i)} \cdot v_y, \quad (3)$$

where $a_x^{(i)}$ and $a_y^{(i)}$ are coefficients determined solely by the positions of the transceivers:

$$\begin{aligned} a_x^{(i)} &= \left(\frac{x_t^{(i)}}{\|\mathbf{l}_t^{(i)}\|_2} + \frac{x_r^{(i)}}{\|\mathbf{l}_r^{(i)}\|_2} \right) = (\cos\theta_t + \cos\theta_r), \\ a_y^{(i)} &= \left(\frac{y_t^{(i)}}{\|\mathbf{l}_t^{(i)}\|_2} + \frac{y_r^{(i)}}{\|\mathbf{l}_r^{(i)}\|_2} \right) = (\sin\theta_t + \sin\theta_r), \end{aligned} \quad (4)$$

where θ is the angle between the normal of the ellipse with focus on the i -th transmitter and receiver and the x axis, and the normal orientation of the target position is $\mathbf{d}^{(i)} = (-a_y^{(i)}, a_x^{(i)})$.

Therefore, we can obtain the velocity components v^i of link i on both x-axis and y-axis respectively:

$$\begin{aligned} v_x^{(i)} &= \cos\left(\frac{a_x^{(i)}}{\|\mathbf{d}^{(i)}\|_2}\right) \cdot v^{(i)}, \\ v_y^{(i)} &= \sin\left(\frac{-a_y^{(i)}}{\|\mathbf{d}^{(i)}\|_2}\right) \cdot v^{(i)}, \end{aligned} \quad (5)$$

only $v^{(i)}$ varies with the movement of the target. For two transceiver links i, j in this experiment, if there exist two velocities $\mathbf{v}$ and $\mathbf{q}$ with the same magnitude but different orientations, where $\mathbf{v} = (v_x, v_y)$ and $\mathbf{q} = (q_x, q_y)$, then we can obtain the following formula:

$$v^{(i)} - q^{(i)} = a_x^{(i)} \cdot (v_x - q_x) + a_y^{(i)} \cdot (v_y - q_y), \quad (6a)$$

$$v^{(j)} - q^{(j)} = a_x^{(j)} \cdot (v_x - q_x) + a_y^{(j)} \cdot (v_y - q_y), \quad (6b)$$

where $v^{(i)}$ and $q^{(i)}$ represent the velocity components of $\mathbf{v}$ and $\mathbf{q}$ on link i , respectively. It can be deduced that $(v^{(i)} - q^{(i)}) + (v^{(j)} - q^{(j)}) = \varepsilon$, where ε is a very small error.

By adding Eq. (6a) and Eq. (6b), which approximate to 0, we can obtain the final result:

$$\begin{aligned} (v_x^{(i)} + v_x^{(j)}) - (q_x^{(i)} + q_x^{(j)}) &= \cos\left(\frac{a_x^{(i)}}{\|\mathbf{d}^{(i)}\|_2}\right) \cdot (v^{(i)} - q^{(i)}) \\ &\quad + \cos\left(\frac{a_x^{(j)}}{\|\mathbf{d}^{(j)}\|_2}\right) \cdot (v^{(j)} - q^{(j)}) \\ &= \sigma, \\ (v_y^{(i)} + v_y^{(j)}) - (q_y^{(i)} + q_y^{(j)}) &= \sin\left(\frac{-a_y^{(i)}}{\|\mathbf{d}^{(i)}\|_2}\right) \cdot (v^{(i)} - q^{(i)}) \\ &\quad + \sin\left(\frac{-a_y^{(j)}}{\|\mathbf{d}^{(j)}\|_2}\right) \cdot (v^{(j)} - q^{(j)}) \\ &= \sigma', \end{aligned} \quad (7)$$

where σ is an error factor smaller than ε , which can be ignored in the experiment.

Finally, we can observe that when two velocity components with different orientations are superimposed on the coordinate axes using two WiFi transceiver links, they can have the same magnitude. Therefore, we extract domain-independent features of gestures. Finally, all K paths are mapped onto the final FMV features over all time T . We assume that the position of the receiver, transmitter and user performing gestures is known in advance. Please note that the user's position can be obtained through other human tracking methods and will not be described in detail in this paper.

Fig. 6 shows a single FMV of all paths at time t , where Fig. 6a is the FMV for single gesture, Fig. 6b is the FMV for two gestures, and Fig. 6c is the FMV for multiple gestures. Therefore, we can see that our FMV model can extract motion information from super-gestures. The final FMV features are denoted as $FMV_{v_x \times v_y \times T_s}$. We have implemented cross-domain feature extraction for super-gesture and inputted the FMV features into subsequent gesture classification module.

V. GESTURE CLASSIFICATION

With the extracted features, we design a hybrid neural network for gesture classification. To ensure accuracy, we begin by normalizing the data to eliminate any negative impact from outlier values. The normalized features are then inputted into the hybrid neural network for gesture recognition.

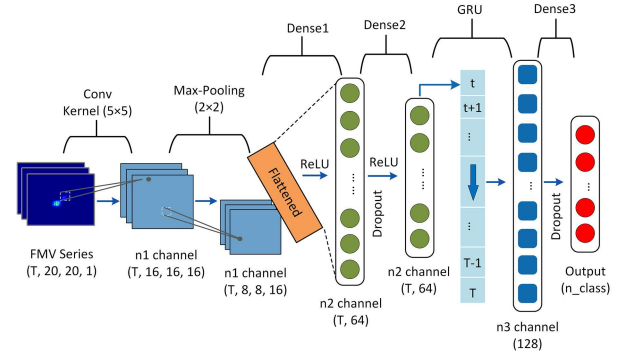


Fig. 7: CNN-GRU neural network architecture.

A. FMV Normalization.

Conceptually, FMV features remove domain-related factors, while possessing the ability to recognize super-gesture. However, in fact, unstable practical factors may cause FMV features to indicate gesture errors. Specifically, when exporting FMV features, we combine the power values of multiple transceiver links. Each transceiver link has multiple body parts that cause the same frequency shift f_{ds} at time t , as well as multiple body parts that cause different f_{ds} . In the former case, as described in Section IV-A, we perform power summation to eliminate this issue and ultimately make the FMV features exported from multiple transceiver links unrelated. Therefore, before inputting FMV features into subsequent deep learning models, we normalize each moment's FMV.

B. Hybrid Neural Network Model based on CNN and GRU.

Although the CNN-GRU model does not have cross-domain capability, it can achieve high-precision cross-domain recognition accuracy when FMV features are used as input. The hybrid neural network of CNN and RNN is shown in Fig. 7. The normalized FMV features have a dimension of $N \times N \times T_s$. We first use a convolutional layer with a max pooling layer to highlight local feature representation, then input it into GRU to extract feature information for all time segments T_s , and finally use fully connected layers for gesture classification. Specifically, for the t -th single FMV, the $FMV_{N \times N}$ matrix is sequentially fed into a 3×3 convolutional layer and two 64-unit fully connected layers. A Dropout layer is added between the two fully connected layers to prevent overfitting. ReLU activation function and Flatten layer are used for non-linear mapping and transition from convolutional layer to fully connected layer, respectively, resulting in the CNN output of the t -th single FMV. Then, the time series of CNN output features are sequentially fed into a 128-unit single-layer GRU and Dropout layer. Finally, Softmax activation function with cross entropy loss function is used to classify gestures.

VI. EVALUATION

Our approach for multi-person gesture recognition can also be applied to single-person scenarios by removing the MMT decomposition module and keeping all other components. We evaluate our approach for both scenarios in this section.

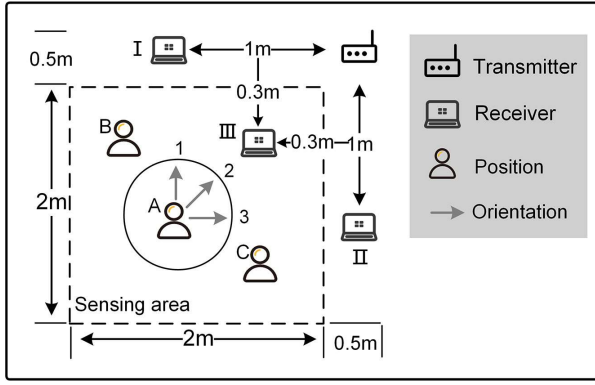


Fig. 8: Our Experimental Environment.

A. Experiment Methodology

Implementation. The WiSG system comprises a signal transmitter, two signal receivers, and an algorithm processor. All the transmitting and receiving devices are running on Ubuntu 14.04 with kernel version 4.2 and equipped with Intel 5300 NICs. We use 802.11n CSI Tool [26] for signal acquisition, where the transmitter activates one antenna while all receivers activate three antennas each. The devices operate in monitor mode on CH-165 at 5.825GHz because this setting allows for precise control over packet size, packet interval, and other parameters that affect CSI measurement results. Furthermore, due to the fact that signal interference has a significant impact on the measurement results of CSI signals in smaller channels, we chose CH-165 for less interference. The transmitter sends data packets at 1000Hz interval. The processor, equipped with an NVIDIA 3090 24GB GPU, processes CSI data through MATLAB and Keras to implement the entire system.

Evaluation Setup. To evaluate the performance of gesture recognition, we build a Yo-Yo platform for e-learning, which is able to capture the challenges of multi-person recognition and cross-domain recognition, in practice, and yield performance metrics to be evaluated. Our WiSG system is deployed on the Yo-Yo platform, enabling students to perform gesture recognition without any human-computer interaction in the classroom. Fig. 8 shows the layout of the sensing area in the classroom. During the experiment, we ensure that there are no other interfering activities. For setting up the sensing area, users may perform gestures at three positions (pos-A, pos-B, pos-C) and towards three orientations (ori-1, ori-2, ori-3). Position factors and orientation factors are our main indicators for evaluating cross-domain performance in this experiment. We set the range of the entire sensing area as $2m \times 2m$. Please note that a $2m \times 2m$ square is a typical indoor interactive gesture recognition setting. All users performed their respective gestures while sitting on chairs and executing their gestures at a height higher than that of desks (75cm). The receiver and transmitter are both deployed on top of desks at a height level with user's gestures.

Dataset. Our implementation targets e-learning environment, but currently available open-source datasets do not meet our

TABLE I: Configuration of Single-&Multiple- Person

Single-person	SET	1	2	3	4	5	6
	-	user1	user2	user3	user4	user5	user7
Two-person	SET	7	8	9	10	11	12
	pos-B	user2	user3	user4	user3	user6	user5
	pos-C	user3	user4	user2	user6	user5	user3

TABLE II: Configuration of Super-gesture

ID	1	2	3	4	5	6	7	8	9	10
#Sub-ges.	1	1	1	1	2	2	2	2	2	2
Sub-ges.	H	W	C	Z	H+W	H+C	H+Z	W+C	W+Z	C+Z

requirements. Therefore, we build our own data collection system in a real-world environment to gather data that is tailored for e-learning classrooms. The deployment of sensing areas is shown in Fig. 8, with a total of three positions (referred to as pos-A, pos-B and pos-C) and three orientations (referred to as ori-1, ori-2 and ori-3). We select four commonly used gestures in classrooms: *raising/lowering hand* (H), *waving hand* (W), *clapping hands* (C), *drawing zigzag* (Z).

Seven volunteers with different heights and body types participated in our experiment. We divide the data collection part into two categories: single-person gestures and two-person super-gestures. Table I shows the configuration of multiple persons, where pos-A and pos-B represent the users who are positioned there and perform gestures. SET-1 to SET-6 represent single-person scenario, while SET-7 to SET-12 represent two-person scenario.

Table II presents the list of gestures performed by humans. Super-gesture ID 1-4 are single-person scenario, while Super-gesture ID 5-10 are two-person scenario. In order to capture two-person gestures, two different users perform different gestures, simultaneously. This is because similar hand movements can occur with the same gesture type, making it difficult to distinguish between them. Therefore, we combine the four single-person gestures in pairs to ensure that each gesture in the multi-person category was performed three times.

In order to reduce the impact of irregular gestures on recognition results, we play gesture execution animations for each volunteer during their first gesture collection session, so that they can clearly understand the correct way to execute the gestures. The duration of one gesture collection is 30 seconds, with a sample recorded every 3 seconds, resulting in 10 samples per collection. We use a prompt tone to inform volunteers when to start executing the gestures. Considering that frequent execution of gestures can cause fatigue among volunteers and lead to significant changes in their performance before and after fatigue sets in, which will affect data collection results. Therefore, we limit each volunteer's collection time to 30 minutes and conduct another round of collections after a certain period of rest in order to minimize the impact of fatigue on data collection.

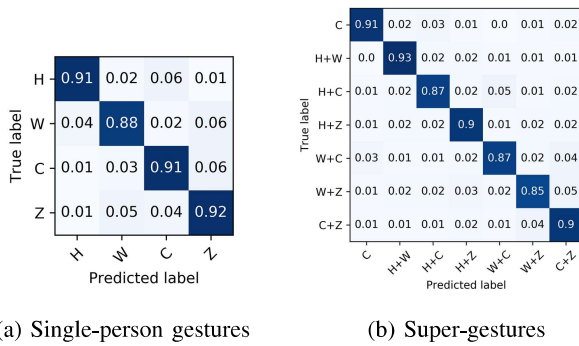


Fig. 9: Average recognition results of gestures.

B. Overall Performance

In all performance evaluations of the experiments, we divide gestures into two categories: single-person gestures (SG ID 1-4) and super-gestures (SG ID 3, 5-10), where SG ID 3 is a clapping gesture that belongs to both single-person gestures and super-gestures. We first evaluate the overall cross-domain gesture recognition accuracy, considering all factors in in-domain and cross-domain situations. The average recognition accuracy of single-person gestures is above 90%, while that of super-gestures is 89%. We achieve consistent high precision for the ten sub-gestures and super-gestures. Fig. 9 shows the recognition accuracy for different gestures. From the experimental results, it can be observed that the super-gestures 'H+C', 'W+C', and 'W+Z' correspond to low accuracy. This is because, during a certain period of time in the execution process of these super-gestures, multiple arms with similar movements produce similar motion trajectories. As a result, some motion information may be lost when extracting MMT features, leading to reduced recognition accuracy.

C. Cross-domain Evaluation

This section evaluates the cross-domain performance of gestures, which mainly covers two aspects: single-person gesture cross-domain experiment and super-gesture cross-domain experiment. To evaluate the influence of domain factors on gesture recognition performance, we set experiments where each target label is treated as the target domain, and the remaining labels are considered as the source domain.

Table III shows the overall cross-domain performance. For single-person gestures, the results show that in the cross-position experiment, the cross-domain performance at pos-A is the best, while the cross-domain accuracy at pos-B and pos-C has decreased. We speculate that this may be because other parts of the body obstructed the gesture, resulting in incomplete gesture motion information received by the transceiver link. For example, when a user performs a gesture towards ori-1 at pos-B, part of the gesture motion information will be missing from CSI signal received by transceiver link-II. In cross-orientation experiments, ori-1 has better recognition performance than others with highest accuracy achieved across all three orientations. The average recognition accuracy for each orientation is above 90%, proving our proposed

TABLE III: Single-person gesture cross-domain performance (in-domain: above 95%) and super-gesture cross-domain performance (in-domain: 89%)

Target label	Cross position			Cross orientation		
	A	B	C	1	2	3
Single-person gesture	89.20%	86.36%	82.41%	93.70%	92.48%	90.96%
Super-gesture	-	-	-	88.71%	86.71%	87.57%

Target SET	Cross user					
	1	2	3	4	5	6
Single-person gesture	90.59%	86.91%	88.66%	88.61%	78.33%	88.41%
Target SET	7	8	9	10	11	12
Super-gesture	86.92%	86.28%	84.47%	89.05%	84.31%	88.77%

method's effectiveness in recognizing single-person gestures across different orientations. In cross-user experiments, we mainly consider how different users' habits affect recognition performance when performing gestures. SET-1 achieves the overall optimal performance, while SET-5 exhibits lower overall cross-domain accuracy compared to other users. This difference can be attributed to variations in body size and habitual execution of gestures, which results in insufficient sample coverage for their specific gesture features.

For the super-gesture, it can be seen from the results that in cross-orientation experiments, the average recognition accuracy of each orientation is above 85%. Although compared to the recognition accuracy of single-person gestures, there is a slight decrease, it still reaches an acceptable level of performance. This further demonstrates the effectiveness of our method in dealing with cross-orientation recognition problems for super-gesture. In cross-user experiments, the average recognition accuracy of each setting is above 84%, and SET-10 has the best performance in cross-user testing. The overlaying of gesture differences between multiple users has little effect on our experiment.

D. Comparison to Other Methods

We compare our method with the existing cross-domain gesture recognition methods EI [22] and Widar3.0 [18]. It should be noted that the best recognition performance of BVP features is achieved with six receivers, while we only use three receivers, so we compare the BVP features and our method under three receivers. We evaluate the cross-domain performance of different methods for single-person gestures, as shown in Fig. 10a. Under single-person gestures, our proposed method has better cross-domain performance than EI and Widar3.0, and when crossing different domain environments, our method has high recognition accuracy.

Fig. 10b shows the results of cross-domain comparison for super-gestures, demonstrating a significant improvement over EI and Widar3.0 achieved by our proposed method. It is worth noting that to our knowledge, we are the first to successfully implement cross-domain recognition of super-gesture, while EI and Widar3.0 do not perform well in dealing with such cases. Specifically, the median value of our method's accuracy is more than 10% higher than that of Widar3.0's median value. This indicates that our proposed method has better overall

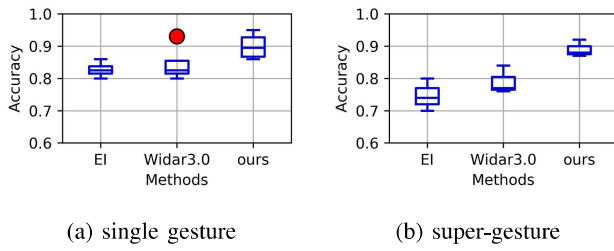


Fig. 10: Cross-domain comparison with other methods.

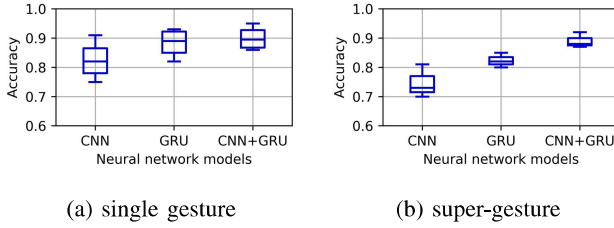


Fig. 12: Cross-domain comparison of different DNN.

performance in dealing with cross-domain situations involving super-gestures than current methods.

E. Parameters Analysis

Different Input Features. We evaluate four different input features to verify the effectiveness of our method, including DFS power spectrum features, MMT features, FMV features without MMT decomposition, and FMV features. We evaluate the cross-domain performance of different input features for single-person gesture, including in-domain, cross-position, cross-orientation and cross-user domains. The experimental results are shown in Fig. 11. It can be observed that our method achieves an accuracy improvement of over 10% compared to the DFS power spectrum feature in the case of single-person gestures. Furthermore, the FMV feature after MMT decomposition exhibits the best performance in cross-domain gesture recognition. These experimental results indicate that our proposed feature extraction method is effective under single-person gesture conditions. Under super-gesture conditions, compared with DFS power spectrum method, our method has significantly improved accuracy by more than 25%. At the same time, we find that the cross-domain recognition performance of FMV without MMT feature decomposition is poor while that with MMT feature decomposition is good, which proves that our MMT feature can extract motion characteristics of super-gesture. Furthermore, the difference between the cross-domain recognition accuracy of FMV and MMT features demonstrates the effectiveness of our proposed FMV model when facing super-gesture situations.

Different Network Structures. We conduct a comparative analysis using CNN and RNN, respectively. The network structures of these two types of networks are the same as the corresponding parts used in this paper's combination network based on CNN and GRU. As shown in Fig. 12, we can see that the cross-domain recognition performance of a single neural

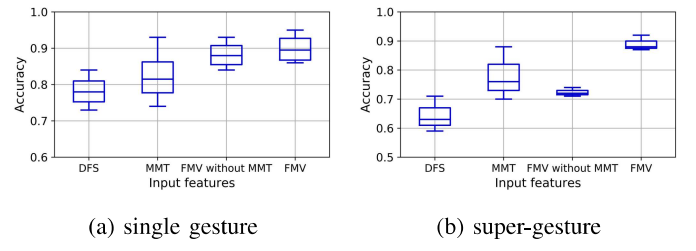


Fig. 11: Cross-domain comparison of different input features.

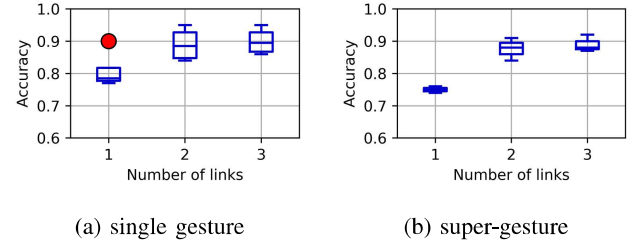


Fig. 13: Cross-domain comparison of different transceiver links.

network structure is inferior to that of a combined neural network. This is because the combined neural network can take advantage of both CNN and GRU, *i.e.*, extract spatial features and temporal continuity features from data, simultaneously. In the case of single-person gesture and super-gesture, using only CNN or only GRU results in lower recognition accuracy than using them together in our proposed FMV model, which already has spatial features but requires more emphasis on time series analysis. Therefore, employing solely GRU yields satisfactory results, whereas combining CNN and GRU in a hybrid network attains optimal performance.

Different Number of Transceiver Links. In our experiments, we deploy one transmitter and three receivers with a total of three transceiver links to accurately estimate the FMV model. The focus of this section is to explore the impact of the number of transceiver links on system recognition performance. As shown in Fig. 13, we find that decreasing the number of transceiver links from three to two has no significant impact on recognition accuracy. However, when the number is further reduced to one, there is a significant decrease in accuracy. This is due to the fact that, as discussed in Section IV-B, the FMV model relies on multiple transceiver links. When there is only one link available, the FMV model is unable to accurately capture domain-independent features of gestures, leading to a decrease in recognition accuracy.

Different Numbers of Arms in Gestures. In order to accurately evaluate the recognition performance of different types of hyper-gestures in this paper, we divide the gestures into different categories, including single-person single-arm gestures, single-person dual-arm hyper-gestures, two-person dual-arm hyper-gestures and two-person triple-arm hyper-gestures. We explore the influence of different numbers of arm movements on system recognition performance, and the results are shown in Fig. 14. The results show that in dual-arm hyper-gestures, the accuracy of recognition by two people is slightly lower

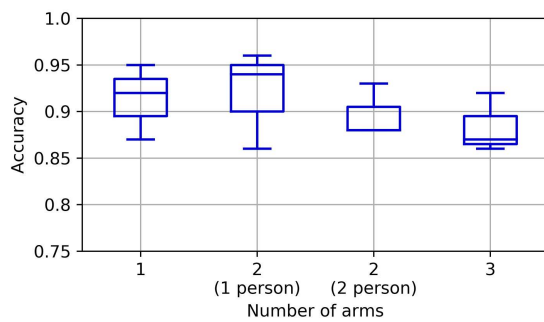


Fig. 14: Cross-domain comparison of the number of arm movements.

than that by one person. While in triple-arm hyper-gestures, the accuracy of recognition is slightly lower than that in dual-arm hyper-gestures. However, these decreases are within an acceptable range and are considered normal.

VII. CONCLUSION

In this paper, we propose a method for WiFi-based multi-person gesture recognition. Particularly, we define the concept of super-gesture, and perform concurrent gesture recognition via the decomposition of the super-gesture's DFS spectrogram. Furthermore, to mitigate the cross-domain effect, we fully exploit the advantages of our multiple transmitter-receiver WiFi sensing system, and propose a feature that is not significantly influenced by domains. Evaluation results show that our WiSG achieves above 90% cross-domain recognition accuracy for single-person scenarios, which is comparable to the state-of-the-art single-person solution (*i.e.*, Widar3.0). For the multi-person scenario, our cross-domain recognition accuracy is 89%, which has a 10% improvement against the baseline that directly adapts Widar3.0 to multi-person scenarios. Besides, our approach can reduce inference time by 20 times against Widar3.0, which satisfies the requirements of most real-time applications.

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